

Satellite Intelligence Canvas — Tools & Utilities Inventory (bottom → top)

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1. Scope and method

This report inventories every tool, layer class and utility that participates in the **Satellite Intelligence** map canvas (route `/satellite`, page component `frontend/src/pages/satellite/SatelliteIntelligence.tsx`, ~29k lines) and the panels mounted around it. It was produced by static reading of the source tree only — no environment was created, no dependencies installed, nothing executed.

The canvas is a single **Mapbox GL JS** map (`react-map-gl/mapbox` `<MapGL>`, globe projection). Part I walks the render stack from the bottom-most pixel (terrain mesh and basemap rasters) to the top-most DOM overlay (toolbox chrome, swipe handle, remote-sensing panels). Part II catalogues each tool family in detail. Part III lists libraries, backend services and a file index.

Conventions:

- `monospace` → Mapbox source/layer id, TypeScript symbol or file path.
- Paths are relative to `frontend/src/` unless they begin with `backend/` or `services/`.
- "SI" = Satellite Intelligence.

2. PART I — Canvas stack, bottom to top

The stack below is the **effective** draw order. Some tiers are declared as React `<Source>/<Layer>` children of `<MapGL>`; others are added imperatively (`map.addSource/addLayer`) and then re-ordered by two z-order utilities:

- `lib/siCustomLayerZOrder.ts` — `siParkBasemapRastersBelowOverlays()` slides every basemap raster immediately under the first non-basemap layer; `siRaiseCustomLayersAboveBasemap()`

raises imported GIS overlays (`-fill/-line/-circle/-marker/-arcgis-icon/-cluster/-cluster-count/-raster/-arcgis-raster/-label`, `-vt-*`, `-arcgis-*`) above them and keeps sketch chrome on top.

- `lib/siMapAnalysisLayerOrder.ts` — `syncSiMapAnalysisLayerOrder()` enforces **basemap → AOI fill → analysis raster → vector outlines → draft sketch → MCDA**.
- Both are invoked from `raiseOverlaysThenAnalysisOrder()` (SI page, ~L19524) on `onStyleData`, after custom-layer paint, and when the Layers-AOI raster stack is (re)mounted.

2.1. Tier 0 — Rendering engine, projection, sky

Item	Implementation	Notes
Map engine	<code>mapbox-gl 3.18</code> via <code>react-map-gl 8</code> (<code>MapGL</code> , <code>Source</code> , <code>Layer</code> , <code>Marker</code>)	<code>reuseMaps</code> , <code>antialias=false</code> , <code>attributionControl=false</code> , <code>dragRotate=false</code>
Projection	<code>projection={{ name: 'globe' }}</code>	<code>minZoom=0.35</code>
Globe cockpit	<code>pages/satellite/siGlobeCockpit.ts</code>	<code>SI_GLOBE_COCKPIT_2D_VIEW</code> , fog preset <code>SI_GLOBE_COCKPIT_FOG</code> , auto-spin (<code>bindSiGlobeCockpitSpin</code> , 300 s), max zoom 2.75 / pitch 12 while in cockpit
Sky backdrop	<code>siGlobeSkyGl.ts</code> , <code>siGlobeSkySurvey.ts</code> , <code>siGlobeSkyCatalog.ts</code>	STScI DSS + SDSS HiPS survey imagery, bright-star / constellation line overlay
WebGL / globe resilience	<code>lib/siMapboxGlobeCompat.ts</code>	Edge detection, WebGL-context-loss recovery with back-off, error classification
Navigation props	<code>lib/agroCloudMapNavigation.ts</code> (<code>AGRO_CLOUD_MAPBOX_NAVIGATION_PROPS</code> , <code>ensureAgroCloudMapScrollZoom</code> , orbit drag)	Wheel zoom, right-drag orbit; no compass/zoom widgets
Camera helpers	<code>lib/googleEarthFlyTo.ts</code> (<code>flyToLikeGoogleEarth</code> , <code>zoomFromLngLatBbox</code>), <code>syncAgroCloudMapboxCamera</code>	Space-lift fly-to
Interaction perf	<code>lib/siMapInteractionPerf.ts</code> , <code>lib/siMapViewport.ts</code> , <code>lib/siViewportFeatureCache.ts</code>	Live viewport extent capture / throttled commits, dense-layer interaction mode
Branding	<code>applyAgroCloudMapboxBranding</code>	Replaces Mapbox logo/attribution chrome

2.2. Tier 1 — Terrain mesh and hillshade (3D basemaps only)

Item	Implementation	Notes
DEM source	<code>lib/agroCloudMapTerrain.ts</code> → raster-dem source <code>esri-terrain</code> from <code>/api/terrain/esri-rgb/{z}/{x}/{y}</code> (Esri <code>WorldElevation3D</code> → <code>Terrain-RGB</code> proxy)	Added at runtime by <code>syncAgroCloudTerrain3d()</code> only after <code>ensureTerrainApiAvailable()</code> ; <code>warmAgroCloudTerrainDemSource()</code> on load

Item	Implementation	Notes
Terra in	<code>map.setTerrain({ source, exaggeration })</code>	Exaggeration slider (<code>setAgroCloudTerrainExaggeration</code>), enabled when pitch > <code>AGRO_CLOUD_TERRAIN_PITCH_THRESHOLD</code> (8°) and basemap is a <code>terrain3d</code> entry
Hills hade	<code>agrocloud-hillshade</code> (computed) and <code>agrocloud-esri-hillshade</code> (Esri World_Hillshade raster overlay)	Part of 3D Topographic style; treated as basemap id by z-order code
2D/3 D toggl e	SI chrome button "Switch to 3D/2D view"	Pitches camera; drives terrain sync in <code>onMoveEnd</code>

2.3. Tier 2 — Basemap rasters

Catalog: `pages/satellite/basemapCatalog.ts` (`buildBasemapCatalog`, shared with the Leaflet GIS Map). All entries are **raster tile** styles built by `rasterStyleFromTiles()` (no Mapbox-hosted vector styles). Layer ids are `layer-0`, `layer-1`, ... inside the style; recognised as basemap by `isSiBasemapStyleLayerId()` (also `topo-base-layer`, `sat3d-base-layer`, `google-earth-sat-layer`, `agrocloud-basemap-layer-*`, `topo-fallback-layer-*`).

id	Label	Tiles
<code>google-earth-satellite</code>	Google Earth Satellite	<code>mt{0-3}.google.com/vt/lyrs=s</code> (needs Google Maps API key; 403 → fallback)
<code>satellite</code> (default)	Satellite (Esri)	<code>Esri World_Imagery</code>
<code>terrain-opentopo</code>	Terrain (OpenTopo)	<code>OpenTopoMap</code>
<code>3d-satellite</code>	Satellite 3D (Esri Imagery + Terrain)	<code>World_Imagery + DEM mesh (terrain3d)</code>
<code>3d-topographic</code>	3D Topographic	<code>World_Topo_Map + Esri World_Hillshade + DEM mesh (terrain3d)</code>
<code>osm</code>	OpenStreetMap	<code>tile.openstreetmap.org</code>
<code>esri</code>	Esri World Imagery	<code>World_Imagery</code>
<code>esri-imagery-hybrid</code>	Imagery Hybrid	<code>World_Imagery + Reference/World_Boundaries_and_Places</code>
<code>esri-streets</code>	Streets	<code>World_Street_Map</code>
<code>esri-topo</code>	Topographic / Outdoor	<code>World_Topo_Map</code>
<code>esri-navigation-night</code>	Streets (Night)	<code>Carto dark_all</code>
<code>esri-terrain-labels</code>	Terrain with labels (Esri)	<code>World_Terrain_Base + Reference/World_Reference_Overlay</code>
<code>esri-light-gray</code>	Light Gray Canvas	<code>Canvas/World_Light_Gray_Base + _Reference</code>
<code>esri-dark-gray</code>	Dark Gray Canvas	<code>Canvas/World_Dark_Gray_Base + _Reference</code>
<code>esri-oceans</code>	Oceans	<code>Ocean/World_Ocean_Base + _Reference</code>
<code>esri-natgeo</code>	National Geographic	<code>NatGeo_World_Map</code>

id	Label	Tiles
esri-shaded-relief	Shaded Relief	World_Shaded_Relief
esri-physical	World Physical	World_Physical_Map
carto-positron	Light (Carto)	Carto light_all
carto-dark-matter	Dark (Carto)	Carto dark_all

Utilities: `resolveBasemapId()` (legacy id aliasing), `pickDefaultBasemapId()`, `basemapTileErrorShouldFallback()` (auto-fallback on 401/403, wired in `MapGL onError` → `tryFallbackBasemapFromTileError`), `isSatelliteImageryBasemapId()`, `getBasemapThumbnail()`, `ESRI_BASEMAP_IDS` picker order, `lib/rasterTileZoom.ts` (ensure `RasterStyleMaxNativeZoom`, provider max-native-zoom table). Picker UI is the inline **Basemap** gallery in the SI left chrome (`BasemapGallery.tsx` is the Leaflet counterpart).

2.4. Tier 3 — AOI polygon fills parked under analysis rasters

`DRAWN_AOI_FILL_UNDER_RASTER_IDS` + `extraFillIdsUnderRaster`:

Layer id	Source	Purpose
drawn-index-geometry-fill	drawn-index-geometry-source (drawn AOI)	Light tint under imagery; opacity 0 while a new shape is being drafted
si-crop-class-aoi-fill	si-crop-class-aoi-source	Crop-classification AOI tint
agro-structures-fs21-fill	Agro_Structures primary AOI layer (<code>lib/agroStructuresPrimaryAoi.ts</code>)	Fill opacity forced to 0 while any Sentinel WMS is on (<code>suppressAgroFillWhenWms</code>)
Mask-builder source layer fill	<code>SiImportedCustomLayersOverlay</code> <code>suppressFillOpacityLayerIds</code>	Outline kept, fill hidden so live WMS owns the fill

2.5. Tier 4 — Analysis / imagery rasters

Analysis rasters are placed just below the top-most `*-line` vector outline (`resolveSiAnalysisRasterBeforeLayerId`).

2.5.1. 4a. Sentinel Hub / CDSE WMS "Layer Live" stacks

Stack	Ids	Mount style	Notes
Drawn-AOI stack	<code>sentinel-draw- <chunk> source / layer</code> (<code>siSentinelAoiWmsSourceId/LayerId</code>)	Declarative <code><Source type="raster">=</code> per display chunk, <code>bounds = chunk</code> <code>bbox</code>	Built by <code>lib/siSentinelAoiWmsStack.ts</code> <code>buildSiSentinelAoiWmsStackState()</code> ; tiles carry WMS GEOMETRY (EPSG:3857 WKT) + base64 EVALSCRIPT (<code>lib/sentinelHubWmsAoiClip.ts</code>) so pixels outside the AOI are transparent

Stack	Ids	Mount style	Notes
Layers-AOI stack	<code>sentinel-layer-aoi-...-s0/-s1</code> ping-pong pairs (<code>lib/siSentinelAoiWmsImperative.ts</code>)	Imperative <code>useLayoutEffect</code> (no React Source)	Two raster sources per chunk; previous frame stays visible until the next tile set loads, then opacity swaps (<code>lib/siSentinelAoiWmsFlickerFree.ts</code>) — zero blank frames on date/layer change
In-place sync	<code>syncSiSentinelAoiWmsChunkTiles/Bounds</code>	<code>setTiles</code> / <code>setBounds</code> directly on <code>RasterTileSource</code>	react-map-gl does not apply standalone <code>bounds</code> updates
Tile geometry	<code>SENTINEL_HUB_WMS_TILE_PIXELS</code> , <code>SENTINEL_HUB_S2_MAX_METERS_PER_PIXEL</code>		

2.5.2. 4b. Other rasters on the canvas

Layer id	Source type	Producer
<code>si-swipe-before-lyr-<i><i></i></code>	raster tiles	MapSwipe Before side (<code>SiMapSwipeRasterLayers</code>) — see swipe section
<code>crop-ai-prediction-raster</code> (<code>SI_CROP_AI_PREDICTION_RASTER_LAYER_ID</code>)	image	Prithvi Crop AI prediction PNG; participates in AOI fill/outline stacking
<code>si-stac-thumb-layer</code>	image	STAC scene thumbnail preview (<code>stacMapThumb</code>)
<code>chirps-precip-raster</code>	image	CHIRPS daily rainfall raster (<code>useChirpsPrecipitation</code>)
<code>collection-dem-index-raster</code>	image	Copernicus-DEM-collection derived index (slope/aspect/hillshade/TPI/TRI/TWI/watershed) from Terrarium tiles (<code>useCollectionDemOverlay</code> , <code>lib/collectionDemOverlay.ts</code>)
<code>hydro-<i><step></i>-raster</code>	image	Hydro Watershed raster steps (DEM, hillshade, slope, flow direction, flow accumulation)
<code>wellsite-heat-raster</code>	image	Well Site Recommendation suitability heat map
<code>segformer-detection-mask-raster</code>	image	SegFormer georeferenced class mask
<code>flood-extent-raster</code> , <code>flood-change-raster</code>	image	SAR flood monitoring
<code>image-classification-raster</code> , <code>-segments</code> , <code>-result</code> , <code>-georef-preview</code>	image	Image Classification wizard
<code>raster-georef-raster</code> , <code>raster-georef-preview</code>	image / raster tiles	Raster georeferencing tool preview & committed layer
<code><layerId>-arcgis-raster</code>	raster tiles	Imported ArcGIS MapServer / ImageServer tile layers (<code>buildArcGisMapServerRasterTiles</code> / <code>buildArcGisImageServerRasterTiles</code>)

Layer id	Source type	Producer
<layerId>-raster	image	Imported GeoTIFF / PNG+world-file / server-ingested raster (renderMode: 'raster', lib/raster/siRasterMapLayer.ts, siRasterTileService.ts ingestRasterFilesViaServer)
ftw-global-* seamless raster ids	image	FTW Global field-boundary "seamless raster" that replaces vector tiles when ready (lib/agriFieldBoundary/ftwGlobalMapLayer.ts)

2.6. Tier 5 — Imported / user vector layers (TOC layers)

Rendered declaratively by components/SiImportedCustomLayersOverlay.tsx so style diffs cannot wipe them; only ArcGIS vector-tile stacks are painted imperatively (siPaintCustomLayersOnMapboxCanvas).

Per-layer Mapbox layers	Notes
<sid>-fill (fill, polygons)	Paint from resolveImportedLayerPaint(): ArcGIS Online drawingInfo (lib/arcgisDrawingInfoMapbox.ts), simple upload style, or Symbology-Studio pack (buildSiCustomVectorStylePack)
<sid>-line (line, polygons + lines)	Zoom-scaled width
<sid>-circle (circle, true point layers only — shouldPaintImportedLayerCircles)	
<sid>-label (symbol)	Field label with SiLayerLabelStyle (font stack, halo, min/max zoom, definition filter)
<sid>-vt-*, <sid>-arcgis-*, <sid>-marker, <sid>-arcgis-icon, <sid>-cluster, <sid>-cluster-count	Imperative ArcGIS vector tiles / point symbols / clusters
ftw-global-pmtiles-source/fill/line	FTW Global PMTiles decoded client-side (@mapbox/vector-tile + pbf + pmtiles, no addProtocol), min zoom FTW_GLOBAL_FIELD_MIN_ZOOM
agro-structures-fs21-*	Agro_Structures FeatureServer layer (Farm Plots / PIVOT codes 1006/1007 mask; outline codes 1000-1002), primary AOI provider

Layer sources (lib/siCustomLayerFactory.ts SiCustomLayerBase.source): 'upload' | 'api' | 'arcgis' | 'stac'; renderMode: 'vector' | 'raster'; geometry kinds 'point' | 'line' | 'polygon' | 'raster' | 'mixed' | 'unknown'; factories createSiVectorImportLayer, createSiRasterImportLayer, createSiBimImportLayer.

2.7. Tier 6 — Tool-result vectors (analysis outputs)

Declared inside <MapGL> in this order (bottom → top):

Source id	Layers	Tool
si-ic-sample-aoi-source	-fill, -line	Image Classification sample AOI

Source id	Layers	Tool
tree-detections-source	tree-detections-overlay (circle), -label (symbol, minzoom 15)	Tree Detections
sam-aoi-source	sam-aoi-fill/line	SAM AOI
sam-train-samples-source	-fill/-line/-point	SAM training samples
training-ai-samples-source	-fill/-line/-point (selected state)	Training & AI samples
sam-train-draft-source	-fill/-line/-point	SAM sample draft
sam-saved-source	-fill, -poly-line, -line, -point	SAM saved objects
sam-mask-source	-fill, -poly-line, -line, -point	SAM current mask
sam-points-source	-halo, -dot (fg green / bg red)	SAM point prompts
agri-field-boundary-source	agri-field-boundary-fill, -outline (tolerance=0, buffer=64)	Field delineation
hydro-<step>-source	-line + -label (contours, colour by elevation), -line (streams, Strahler/Shreve ramp), -fill + -line (mesh/basins/watershed)	Hydro Watershed (persistent until removed in Layers)
si-aoi-fields-source	si-aoi-fields-fill/line	AOI fields (lib/siAoiFields.ts)
si-regional-crop-calibrated	-fill, -line	Regional crop training calibrated output
si-multi-aoi-source	si-multi-aoi-fill (opacity 0), -line (active weight)	Multi-AOI outlines
segformer-detection-source	segformer-detection-fill, -outline	SegFormer polygons
(disabled) si-multi-aoi-centroids cluster, si-aoi-heat-source heatmap		guarded by false &&
si-stac-footprints	-fill, -line (dashed)	STAC search footprints
si-wapi-alert-polygons → si-wapi-alert-fill/line		ISS/WAPI irrigation alerts (SiWapiAlertMapOverlay)
flood-boundaries-vector		SAR flood polygons
wellsite-recommended-wells	managed custom layer	Well points
well-suit-mcda-<ts>-*	heat / streams / points	Well Suitability MCDA — always raised to top by syncSiMapAnalysisLayerOrder
training-ai-live-samples	custom layer	Training results live layer
si-ts-weather-storm	custom layer	Storm periods from imagery-weather analysis
ai-detection-netflora-results, image-classification-samples/-checkpoints, si-gcp-overlay		AI DL detection, IC wizard, georeference GCPs

2.8. Tier 7 — AOI outlines & vertices kept above rasters

`DRAWN_AOI_OUTLINE_ABOVE_RASTER_IDS` + `extraOutlineIdsAboveRaster`:

Layer id	Notes
<code>drawn-index-geometry-line</code> (source <code>drawn-index-geometry-outline-source</code>)	Dashed amber when the AOI comes from a Layers-source AOI
<code>drawn-index-geometry-point</code>	Point AOI marker
<code>si-crop-class-aoi-line</code>	
<code>agro-structures-fs21-line</code> , <code><maskSource>-line</code>	Agro / Layers-AOI outline (insert anchor for rasters)
<code>si-edit-handles-circles</code> (source <code>si-edit-handles</code>)	Vertex edit handles for committed geometry

2.9. Tier 8 — Selection, identify & attention highlights

Source	Layers	Producer
<code>si-geo-ai-table-selection</code>	<code>si-geo-ai-sel-fill/line/point</code>	GIS selection workbench + GeoAI linked table selection (<code>lib/geoAiLinkedSelection.ts</code> , <code>siGeoAiMapSelectionPaint.ts</code>)
<code>si-imagery-field-flash</code>	<code>-fill</code> , <code>-line</code> (blinking)	Time-series → map field flash (<code>siImageryFieldFlashPaint.ts</code>)
<code>si-geo-ai-pin</code>	<code>si-geo-ai-pin-glow</code> , <code>-core</code>	GeoAI answer pin

2.10. Tier 9 — Sketch & measurement chrome (always top of GL stack)

`DRAW_DRAFT_TOP_IDS` / `DRAW_CHROME_ABOVE_OVERLAY_IDS`:

Source	Layers
<code>si-draw-draft</code>	<code>si-draw-draft-fill</code> , <code>-line</code> (dashed), <code>-close-hint</code> (green dashed), <code>-vertex</code> (polyVertex / circleCenter / circleCardinal roles), <code>-pt</code>
<code>si-measure-source</code>	<code>si-measure-fill</code> , <code>-guide</code> , <code>-line-active</code> , <code>-line-done</code> , <code>-vertex</code>

2.11. Tier 10 — HTML markers anchored to the map (`<Marker>` / `SiMapDockAwareMarker`)

Measurement labels · Go-To-XY pin · search pin (pulsing, clickable when a GeoAI card is staged) · GeoAI pin "show details" button · crop-alert field markers + popup (`SiCropAlertMapMarkersLayer`, `SiCropAlertMapMarker`, `SiCropAlertHvdIcon`) · WAPI alert markers + popup · stress-zones popup · multi-AOI popups (source / mean / range) · feature identify popup (`SiFeatureInspectPopup`, overlap stepping) · weather pin. `SiMapDockAwareMarker` keeps popups clear of the right dock.

2.12. Tier 11 — DOM overlays above the canvas (bottom → top in

JSX)

- 1. WeatherVizOverlay — canvas particle weather sim (rain/snow/fog/storm).
- 2. SiCropAlertMapLegend.
- 3. LayerLiveLegendFloatingPanel — index legend + class areas + analyze.
- 4. SiImageryTimeSeriesFloatingPanel.
- 5. SiCropAiFloatingPanel → SiPrithviCropToolPanel.
- 6. SiMapSwipeControl — MapSwipe **After** map (transparent clipped <MapGL>) + divider handle + compare panel.
- 7. GisFloatingWorkspacePanel → SiSymbologyStudioPanel (symbology dialog).
- 8. SiGoToXyBar.
- 9. WeatherIntelligencePanel, WeatherVisualizationPanel.
- 10. SatelliteAoiStaticChartsMapOverlay — on-map multi-layer AOI charts.
- 11. NeighborhoodAgentChat / GeoAI floating widget.
- 12. SatelliteMapAnalysisChrome — left floating controls, right **Map toolbox** dock (SatelliteContextualAnalysisDock → MapToolsDock portal into map.getCanvasContainer()), timeline bar.
- 13. GisSelectionWorkbench, SiMapDrawWidget, HydroLegendTool, draw live HUD, MeasurementPanel.
- 14. SatelliteMapProcessingOptionsPortal — processing panels (Remote sensing, alerts, AI tools, hydro, flood...).
- 15. GeoreferenceFloatbar → RasterGeoreferenceRibbon.
- 16. SiLayerPopupConfigurator, SiLayerLabelingPanel, SiLayerLegendFloatingPanel.
- 17. GisDataManager modal, SiAddArcGisLayerPanel, GisPortalBrowseLayersPanel.

3. PART II — Tool families in detail

3.1. 1. Map chrome & dock controls

3.1.1. Right-rail Map toolbox (SatelliteContextualAnalysisDock.tsx, SatelliteMapAnalysisChrome.tsx, MapToolsDock.tsx)

Primary actions (top → bottom): **Add layer** (add-gis-layer → MapToolboxAddGisLayerFlyout) · **Edit** (toggles AOI drawing, opens SiMapDrawWidget) · **Select** (GisSelectionWorkbench) · **Measure** (MeasurementPanel) · **Go To XY** (SiGoToXyBar) · **Chat AI Agent** (SatelliteGeoAiFloatingWidget) · collapse strip · theme toggle · close panel.

Rail groups (RAIL_GROUPS_MAP):

Grou p	id	Label	Opens
1	layers	Layers	TOC (MapToolboxLayerList)
1	remote-sensing	Remote sensing	RemoteSensingToolboxPanel
1	wapi-alerts	ISS Alert	SiWapiAlertPanel
1	crop-classification	Crop AI	isolated SiCropAiFloatingPanel

Group	id	Label	Opens
1	imagery-time-series	Time Series	SiImageryTimeSeriesFloatingPanel
1	map-swipe	MapSwipe	SiMapSwipeControl
1	layer-live-legend	Legend	LayerLiveLegendFloatingPanel
1	hydro-watershed	Hydro Watershed	HydroWatershedPanel
2	tree-detections	Tree Detections	TreeDetectionsPanel
2	agri-field-boundary	Field Boundaries	AgriFieldBoundaryPanel
2	well-site	Well Site	WellSiteRecommendationPanel
2	well-suitability	Well Suitability	WellSuitabilityPanel
2	flood-monitoring	Flood (SAR)	FloodMonitoringPanel
2	raster-georeference	Georeference	GeoreferenceFloatbar + RasterGeoreferenceRibbon

Other processing-portal section ids reachable by navigation / Smart Processing: crop-alerts (Agro Sentinel Alert Engine → SiCropAlertCenterPanel), stress-zones (SiStressZonesPanel), training-ai (TrainingAITool), segformer-detection (GeoAI Analysis Toolbox → SegFormerDetectionPanel), source (STAC catalog). Retired ids: eo-enrichment, ai-detection-gis, sam-detection, image-classification (panels still exist and mount through other flows). Catalog helpers: SmartProcessingWorkflowPanel.tsx, SatelliteSmartProcessingPanel.tsx.

3.1.2. Left floating chrome (inline in SI page)

Search (places · layers · features; lib/mapSearchGeocode.ts, lib/siMapLayerSearch.ts, lib/geoExplorerGeocode.ts) · Layer Live toggle · Basemap gallery · Weather Intelligence (Open-Meteo pin pick) · Weather visualization · 2D/3D + Terrain relief slider · Key/Legend (HydroLegendTool) · Swipe.

3.1.3. Timeline chrome (SatelliteMapAnalysisChrome.tsx)

Previous / Play-Pause / Next · date chips · scrub nudges · playback-speed cycle.

3.1.4. Not present

No dedicated zoom ±, home, compass, fullscreen or scale-bar widgets on the SI canvas (MapControls.tsx, SearchWidget.tsx, BasemapGallery.tsx, ScaleSelector.tsx, DrawTools.tsx, ToolPanels.tsx serve the Leaflet GisMap.tsx).

3.2. 2. Drawing tools (AOI sketch)

Files: components/SiMapDrawWidget.tsx, components/RemoteSensingDrawingToolbar.tsx, pages/satellite/drawingUtils.ts (DrawStyleConfig, VertexRef, CircleCardinal, geometry helpers), siDrawnAoiLiveIndex.ts (clip signature / normalisation).

Tool id (RemoteSensingDrawingTool)	Behaviour
point	Single click point AOI
circle	Centre + edge; cardinal handles; Enter commits refine; live HUD shows radius/area
rectangle	Drag rectangle
polygon	Click ring; Enter closes (≥ 3 pts); Backspace / Ctrl+Z drops vertex; close-hint segment
Pan / Lock	Re-enable drag-pan or lock the map while sketching
Clear	Clears sketch (analysis layers stay)
×	Turn drawing off

Post-commit editing: vertex handles (`si-edit-handles`), geometry undo/redo, draw style (stroke/fill colour, width, opacity, point radius), AOI layer visibility/opacity in TOC, "Use as AOI" from vector layers, active AOI resolution (`lib/siAoiManager.ts resolveSiActiveAoi` → source `'draw' | 'layers'`).

3.3. 3. Measurement tools

`components/MeasurementPanel.tsx` + `lib/measurement/measurementEngine.ts`. Modes: distance, area, perimeter, rectangle, circle, radius, bearing, angle, coordinate, elevation (3D), distance3d (3D), height3d (3D). Distance units: m, km, ft, mi. Area units: m², ha, acre, km². Actions: undo, redo, finish (double-click), clear current, clear all, close; right-click = undo vertex. Labels rendered as `<Marker>` chips; geometry on `si-measure-source`.

3.4. 4. Selection workbench

`components/gisSelection/*` + `lib/gisSelection/{types,spatialQuery,mapSelectionQuery}.ts`, context `pages/satellite/gisSelection/GisSelectionContext.tsx`.

GisSelectionTool	select (click), rectangle, polygon, lasso, circle, line, trace
GisSelectionSetMode	new, add (Shift), remove (Ctrl), subset "From current" (Ctrl+Shift)

Panels: toolbar, layer list/panel, results (counts, zoom, clear, export), attribute table (`GisSelectionAttributeTablePanel`), **Select by Attributes**, **Select by Location** (intersects, within, contains, overlaps, touches, within_distance km, completely_contains). Engine helpers: `selectFeaturesAtPoint`, `selectFeaturesByMask`, `lineSelectionMask`, `mergeSelectionHits`, `selectFeaturesAtMapPoint` (overlap stack), `computeStableGisFeatureKey`.

3.5. 5. Swipe tool (MapSwipe)

Files: `components/SiMapSwipeControl.tsx`, `components/siMapSwipeTiles.ts`.

- Compares two Sentinel **Layer Live** renderings over the active AOI (disabled without an AOI clip).
- Modes (`SiMapSwipeMode`): `both` (independent layer + date per side), `dates` (same layer, two dates; Before defaults to `-14 d` via `defaultSwipeBeforeDate`), `layers` (same date, two layers).

- **Before** side: tile URLs built by `buildSiMapSwipeTileUrls()` (WMS + AOI clip) and painted on the **main** map as `si-swipe-before-src-<i>/-lyr-<i>` (Tier 4b).
- **After** side: a second, non-interactive `<MapGL>` with a fully transparent style (`SI_MAP_SWIPE_TRANSPARENT_STYLE`) hosting `si-swipe-after-*` rasters, camera-synced to the main map on every `move`, clipped with CSS `clip-path: inset(0 0 0 <split>%)` and a draggable divider (`← / →` nudge $\pm 2\%$).
- While open, the Layers-AOI ping-pong stack is hidden (`layerVisible = ... && !mapSwipeOpen`) and per-side colour-key legends come from `LayerLiveLegendPanel`; the Layer Live legend panel receives `mapSwipeCompare`.
- Inputs: cloud coverage, layer catalogue (`remoteSensingLayerOptions`), scene date (auto-follow or manual).

3.6. 6. Layer list / TOC and layer utilities

`components/MapToolboxLayerList.tsx` rows: visibility eye, drag re-order, opacity slider, zoom-to, attribute table, symbology, properties/metadata, duplicate, rename, AI analysis, generate report, show/hide labels, export raster (GeoTIFF), remove, copy name. SI ... menu (`SiLayerOptionsMenuPortal` in SI page): Zoom to · Layer properties · Export (Shp / KML / GeoJSON / CSV) · Attributes · Symbology · Legend · Configure pop-ups · Layer opacity · Bring forward / Send backward · Copy name · Use as AOI · Rename · Refresh (ArcGIS) · Labeling... · Definition query... · Remove from map. Supporting libs:

`lib/siLayerPopupConfig.ts`, `lib/siLayerPopupInspect.ts`, `lib/siMapFeatureIdentify.ts`,
`lib/siLayerLabelStyle.ts`, `lib/layerLiveLegendCatalog.ts`, `lib/siLayerExport.ts`,
`lib/vectorLayerExport.ts`, `lib/gisMapLayerStore.ts` (saved maps),
`components/GisMapSaveOpenPanel.tsx`, `components/SiDynamicMapSnapshotsPanel.tsx`.

3.7. 7. Add-layer & data import

3.7.1. Add-layer flyout (`MapToolboxAddGisLayerAction`)

`browse` (Browse layers → `GisPortalBrowseLayersPanel` / `GisMapBrowseLayersPane`) · `arcgis` (Add ArcGIS Layer → `SiAddArcGisLayerPanel`, `lib/arcgisServiceDiscover.ts`, `arcgisDynamicLayer.ts`, `arcgisFeatureLayerGeoJson.ts`, `arcgisImageServer.ts`, `arcgisPortalToken.ts`) · `url` (Add layer from URL, `parseRemoteUrlAsFile`) · `file` (Add layer from file) · `sketch` (Create sketch layer) · `media` (Add media layer).

3.7.2. GIS Data Manager (`pages/satellite/gisDataManager/`)

Categories (`GisDataManagerCategory`): `recent`, `favorites`, `giscontent`, `vector`, `raster`, `database`, `cloud`, `web`, `bim`, `lidar`, `realtime`, `ai`, `enterprise`. Wizard: source → preview → projection → symbology → validation → import → ready.

3.7.3. Supported file formats (`utils/FileLoader.ts`)

- Vector: `.geojson` `.json` `.topojson` `.zip` (shapefile) `.kml` `.kmz` `.gpx` `.csv` `.xlsx` `.xls` (+ `.shp` `.dbf` `.shx` `.prj` `.cpg` inside zip) — parsers `shpjs`, `@tmcw/togeojson`, `jszip`, `papaparse`, `xlsx`, TopoJSON converter.
- Raster: `.tif` `.tiff` `.geotiff` `.jp2` `.j2k` `.png` `.jpg` `.jpeg` `.webp` `.gif` `.bmp` + world files `.tfw` `.pgw` `.jgw` `.jpgw` `.wld`, `.prj`, `.aux.xml` — `geotiff`, `proj4`,

`lib/aiDetection/siAiDlRasterPipeline.ts` (CRS detection, bounds reprojection),
`lib/raster/siRasterMapLayer.ts` (8192 px texture cap, fit within bounds).

- BIM: `.ifc` (`createSiBimImportLayer`).

3.7.4. Portal / hosted layers

`lib/gisContentPortalStore.ts`, `gisContentPortalPublish.ts`, `gisHostedFeatureLayerPortal.ts`,
`gisContentPortalTableUtils.ts`, `pages/master/gisContentPortalData.ts`.

3.8. 8. Remote sensing tools

3.8.1. 8.1 Remote Sensing toolbox panel (`components/RemoteSensingToolboxPanel.tsx`)

Provider / collection / layer select (`RemoteSensingLayerSelect`, `SiRsPanelSelect`), AOI mode (draw vs Layers AOI → `SiAoiLayerModePanel`, `SiAoiMaskBuilderPanel`), "Show on map" toggle, imagery date + auto-follow, cloud-cover slider, time-series window, legend toggle, Generate/Stop timeline, Export GeoTIFF, CHIRPS controls, field-timeline CTA.

3.8.2. 8.2 Providers (`lib/remoteSensingProviders.ts`, `remoteSensingProviderSession.ts`, `remoteSensingLayerProfiles.ts`)

id	Label	Backend	Status
<code>sentinel-hub</code>	Sentinel Hub (S2 L2A/L1C, S1 GRD, S3, S5P, S6, CCM, Copernicus DEM, Mosaics, CLMS, Complementary)	<code>services.sentinel-hub.com/ogc/wms/{instance}</code> (<code>lib/sentinelHubWmsInstance.ts</code> , <code>token</code> <code>sentinelHubAccessToken.ts</code>)	integrated
<code>esa-sentinel</code>	ESA Sentinel (CDSE)	<code>sh.dataspace.copernicus.eu/ogc/wms/{id}</code> with SH fallback	integrated
<code>nasa-landsat</code>	NASA Landsat 8/9, HLS	Sentinel Hub	integrated
<code>aster</code>	ASTER L1T	PC STAC calendar, formulas in <code>asterL1tIndices.ts</code>	partial
<code>planet</code> , <code>airbus</code> , <code>maxar</code> , <code>blacksky</code> , <code>satellogic</code> , <code>space42</code> , <code>umbra</code> , <code>mbrsc</code> , <code>jaea</code> , <code>oneatlas</code>	Commercial VHR / SAR	Open S2 placeholder until credentials	not integrated

Layer profiles: `s2-optical`, `landsat-optical`, `s3-optical`, `vhr-optical`, `sar`, `aster-optical`, `collection-catalog`. Other imagery paths: Microsoft Planetary Computer STAC (`lib/siSentinelLatestScene.ts`, `mpcPlanetaryApi.ts` templates `ndvi_s2`, `false_color_s2`, `ndmi_s2`, `ndvi_landsat`, `false_color_landsat`), proxied STAC search (`SentinelSearch.tsx` → `/api/stac/search`; visualisations Natural Color, Color Infrared, SWIR, Agriculture, NDWI, NDMI, EVI, EVI2, SAVI, BSI, NBR, Atmospheric Penetration), Esri ImageServer (`EsriImageServerLayer.tsx`), UCSB CHIRPS (`/api/chirps/*`), SEN2SR super-resolution (`Sen2srProductControls.tsx`, `useSen2srControls.ts` → `/api/sentinel2/super-resolution`; modes RAW S2 10 m · SEN2SR 2.5 m · Basemap RGB · Drone upload).

3.8.3. 8.3 Indices and products

- Core S2 evalscripts (`lib/agroCompositeIndices.ts`, `sentinelHubWmsLayers.ts`, legends in `layerLiveLegendCatalog.ts`): NDVI, NDMI, NDII, NDWI, MNDWI, AWEI, NBR, SAVI, EVI, GNDVI, NDRE, NDSI, ET proxy, LST proxy, DATAMASK; presets `1_TRUE_COLOR`, `2_FALSE_COLOR`, `3_NDVI`; GetCapabilities merge for SWIR / SAR VV-VH.
- Live analysis products: LULC (`lib/siLulcClassification.ts`), ADI (`adiIndex.ts`), NCADI (`ncadiIndex.ts`), CHAS_ALERT, STRESS_ZONES, CROP_CLASS.
- Agro composites (+ Δ deltas): Vegetation Health (CVHI, VHS, VDI, CVI, CSI, WST) · Water & Moisture (DRI, VMI, SMI, OIR, WDSI, DSI) · Irrigation & Field (IEI, UII, FPR, CPI, ISS, WAPI — `wapiIndex.ts`) · Growth & Stability (GPI, CSI2, CRI, VDG) · Risk (ARI, CHS, CPS) · Phenology (PRI, CGI, CMI, HRI, VRI, CCI, EPD, EHD) · Soil & Salinity (NDSI, SI, SSI) · Gold exploration (IOI, CLAY_MI, FMI, NDAI, BSI, REAI, GEI, GCI, EGCI) · CHAS.
- Mangrove (`mangroveIndices.ts`): MVI, REMI, MI, MFI, NDRE-B5/B6/B7, CI-RE, GCI-CHL, MTCI, REIP.
- Per-collection catalogs (`lib/collectionIndexCatalog.ts`): Sentinel-3 (NDVI, EVI, FAPAR, LAI, FCOVER, CI, LST, WQI) · Sentinel-5P (NO2, SO2, CO, O3, CH4, AI, AQI) · Sentinel-6 (SLA, SSH, OST, SWH, SLT) · CCM optical / SAR · Copernicus DEM (slope, aspect, hillshade, TPI, TRI, TWI, watershed) · Sentinel Mosaics · CLMS bio-geophysical / LULC · Complementary climate.
- ASTER L1T (`asterL1tIndices.ts`): vegetation, thermal (LST, NDIE, TAI), soil, mineral (CI, IOI, FMI, CAI, OHI, SILICA, QI, ...).
- CHIRPS (`lib/chirpsRainfall/*`): PRECIP raster; analytics P, RAI, SPI, RTI, RDI, WAI; aggregation daily/monthly/seasonal/annual.

3.8.4. 8.4 Time & scene selection

Imagery date + auto-follow (`lib/siSentinelImageryDate.ts`), latest-scene resolver with cloud filter and cache (`siSentinelLatestScene.ts`), WMS TIME windows (30 d default; ADI 90, NCADI 60, WAPI 45, LULC/crop 120; Δ = previous $\rightarrow$ current), `ImagerySceneCalendar.tsx`, time-series range (90 d default), timeline playback, temporal MapSwipe.

3.8.5. 8.5 AOI clipping & masking

Drawn-AOI WMS clip (`sentinelHubWmsAoiClip.ts`, `sentinel-draw stack`) · Layers-AOI mode (`SiAoiLayerModePanel`, `lib/siAoiMaskBuilder.ts`, `siAoiLayerModeClipCache.ts`, `sentinel-layer-aoi stack`) · Mask Builder (filtered / selected / whole layer; transparent / clip / dim outside) · flicker-free ping-pong stack · `lib/layersAoiClipGeoJson.ts`, `siCropAoiResolve.ts`, `siCropAoiSource.ts`.

3.8.6. 8.6 Analysis & derived tools

Layer Live legend + class areas (`LayerLiveLegendPanel`, `useLayerClassAreas`, SH Statistical API) · Legend Analyze (`LayerLiveLegendAnalyzePanel`, `layerLegendAnalyzeIndexConfig.ts`, `layerLegendAnalyzeStats.ts`) · DEM index overlay · CHIRPS stats · EO Layer Enrichment (`EoLayerEnrichmentPanel`, zonal means + agronomic heuristics) · Imagery Time Series / multi-layer trends / LULC composition (`SiImageryTimeSeriesPanel`) · Weather vs index (`SiImageryWeatherTab`, `lib/imageryWeatherCompare.ts`, `imageryStormAnalysis.ts`) · legacy templates (`AnalysisPanel.tsx`, `AnalysisTool.tsx`).

3.8.7. 8.7 Exports

Index GeoTIFF ZIP (lib/remoteSensing/rsIndexGeoTiffExport.ts: RGB TIF + Float32 index + companions + prj/tfw + README) · CHIRPS GeoTIFF/CSV/XLSX/HTML · vector KMZ/KML/SHP/GeoJSON/CSV/XLSX (vectorLayerExport.ts, VectorLayerExportPanel) · EO enrichment CSV/GeoJSON/DOCX (lib/eoEnrichmentReport/) · time-series report family (see §12).

3.9. 9. AI detection & segmentation tools

Tool	UI / hook	Model	Backend	Map ids
SAM Detection	SamDetectionPanel, SamTrainingSamplesManager, useSamDetection, useSamTrainingSamples	Meta SAM (samgeovit_h/l/b)	/api/sam-detection/* → backend/services/sam-detection	sam-*
AI DL Detect / Train (GIS panel)	aiDetection/SiAiDetectionGISPanel, SiAiDLDetectObjectsPanel, SiAiDLTrainModelPanel, SiAiDLRasterGeorefDialog	dltk / yolo / tensorflow / pytorch / onnx / custom; Netflora YOLO	/api/raster/*; geoai-inference (Mask R-CNN)	ai-detection-netflora-results
SegFormer Detection	segformerDetection/SegFormerDetectionPanel, useSegFormerDetection	SegFormer-B5 ADE20K (+ SAM2 refine, Temporal Transformer / Prithvi)	/api/segformer-detection, /api/sam2-refinement, /api/temporal-transformer	segformer-detection-*
Training & AI	trainingAi/TrainingAITool (Data, Training, Validation, Inference, Quick dashboard), useTrainingAISamples	Trainable SegFormer; engines segformer / delineate-fbis / yolo-trees	/api/training/*, /api/inference/*, /api/delineate-anything/*	training-ai-samples-*, training-ai-live-samples
Image Classification wizard	imageClassification/ImageClassificationWizardPanel, useImageClassificationWizard	SLIC / Felzenszwalb + supervised/unsupervised, pixel/object	/api/image-classification/*	image-classification-*, si-ic-sample-ai-*
Tree Detections	TreeDetectionsPanel, useTreeDetection, lib/treeDetection/	Ultralytics YOLO tree class (+ species classifier, on-device fallback)	/api/tree-detection/predict	tree-detections-*

Exports: GeoJSON, Shapefile, GPKG, CSV, mask PNG; training packs geojson / sam / yolo / unet / segformer.

3.10. 10. Crop & agronomy intelligence

Tool	UI / lib	Notes
Agricultural Field Delineation	AgriFieldBoundaryPanel, ...TimePanel, ...ValidatePanel, ...ResultsDashboard, ...TrainingSamplesPane, FtwAoiTrainingDashboard, hooks useAgriFieldBoundary, useFieldBoundaryTrainingSamples, useFtwAoiTraining, lib/agriFieldBoundary/	AOI modes draw / layers / viewport / select; models ftw, ftw-inference-s2, delineate-fbis, agricultural-field-delineation, map-rgb; FTW Global PMTiles; /api/agri-field-boundary/*, /api/ftw-training/*
Prithvi Crop AI	SiPrithviCropToolPanel in SiCropAiFloatingPanel, lib/siPrithviCropPipeline.ts	IBM Prithvi multi-temporal; /api/crop-classification/*; output crop-ai-prediction-raster; DOCX + GeoTIFF export (lib/cropClassificationReport/)
Crop Classification (WMS) + Regional training	SiCropClassificationPanel, SiCropClassificationAoiToolbar, SiRegionalCropTrainingPanel, lib/siCropClassification.ts, siRegionalCropTraining.ts	Temporal evalscript CROP_CLASS / LULC; regional calibration layer si-regional-crop-calibrated
Smart Crop Alert Center	SiCropAlertCenterPanel, map markers/popup/legend, lib/siCropAlertEngine.ts, siCropAlertSentinelLive.ts, siCropAlertImageryValidation.ts	Per-field NDVI/NDWI/NDMI/EVI rule + Δ CHAS engine
ISS / WAPI Irrigation Alert	SiWapiAlertPanel, SiWapiAlertMapOverlay, lib/siWapiAlertEngine.ts	Excel export
Stress Zones	SiStressZonesPanel, SiStressZonesMapPopup, useStressZonesAnalysis, lib/siStressZonesLive.ts	WMS STRESS_ZONES + zone areas
Water Status	SiWaterStatusPanel, useFieldWaterRequirement	ET0 / AET / water requirement (Open-Meteo, WaPOR)
EO Layer Enrichment	EoLayerEnrichmentPanel	Attributes or segmentations mode
Field Attributes Dashboard	FieldAttributesDashboard*, lib/objectAttributes/	KPI charts, schema-driven attributes

3.11. 11. Hydrology & water

Tool	UI / hook / lib	Map ids
Hydro Watershed	HydroWatershedPanel, HydroLegendTool, useHydroWatershed, lib/hydroWatershed/hydroEngine.ts, terrainTiles.ts	hydro-<step>-* for dem, hillshade, slope, flow-direction, flow-accum, streams (Strahler/Shreve), contours (+labels), watershed, basins, mesh; GeoTIFF + DOCX report

Tool	UI / hook / lib	Map ids
Well Site Recommendation	WellSiteRecommendationPanel, useWellSiteRecommendation, wellSiteWorkbook.ts, lib/wellSiteReport/	wellsite-heat-raster, wellsite-recommended-wells; GeoTIFF / GeoJSON / CSV / XLSX / DOCX
Well Suitability (MCDA)	WellSuitabilityPanel, useWellSuitabilityAnalysis, wellSuitabilityExports.ts, lib/hydroWatershed/wellSuitabilityMcdaEngine.ts	well-suit-mcda-*; GeoTIFF / KMZ / SHP / PDF / workbook
Flood Monitoring (SAR)	FloodMonitoringPanel, useFloodMonitoring, lib/sarFloodReport/	Sentinel-1 pre/post VV threshold → flood-extent-raster, flood-change-raster, flood-boundaries-vector; DOCX + XLSX

3.12. 12. Weather, climate, time series & reporting

- Weather Intelligence (WeatherIntelligencePanel, WeatherTimeHistoryChart, lib/openMeteoWeather.ts, pages/satellite/lib/weatherClimateReport/) — forecast + archive at pin; DOCX / Excel.
- Weather Visualization (WeatherVisualizationPanel, WeatherVizOverlay, weatherSimModel.ts) — particle sim + Mapbox fog.
- Imagery ↔ weather (imageryStormAnalysis.ts, imageryWeatherCompare.ts, SiImageryWeatherTab) — storm overlay si-ts-weather-storm.
- Time series: hooks/useAoiLiveTimeSeries.ts, useImageryTimeSeriesStream.ts, useMultiLayerAoiTrendStream.ts, usePlotLayerTimeSeriesStream.ts, useImageryIndexInterpretation.ts, utils/aoiLiveTimeSeries.ts, staticAoiMultiChartData.ts, views SiMultiLayerAoiTrendView, SiPlotLayerTimeSeriesView, SiScatterCorrelationInsight, SiImageryIndexInterpretationCard, SatelliteAoiStaticChartsMapOverlay, AoiStaticMultiLayerLineChart (Chart.js).
- Reporting (components/timeSeriesReport/ExportManager.tsx, pages/satellite/lib/timeSeriesReport/*): export kinds pdf, docx, lulc-docx, excel, batch-excel, batch-field-summary, weather-excel, plot-priority-excel, aoi-raw-excel, aoi-raw-by-layer-excel, agri-object-intel-excel, csv, png, geojson (jsPDF + autotable, ExcelJS, DOCX XML builder).
- Map snapshots (SiDynamicMapSnapshotsPanel) → PNG / PDF packs.

3.13. 13. GeoAI assistants (LLM)

UI: geoAiAgent/GeoAiAgentPanel, GeoAiAgentBuilder, neighborhoodAgent/* (chat, composer, evidence rows, stats chart, transcript), GeoExplorerGemini* (chat body / input / message parts), GeoExplorerDynamicTable, GeoAiEditQuestionTool, SiGeoAiInspectPopupBody, SatelliteGeoAiFloatingWidget. Turn runners: lib/runGeoAiServerTurn.ts (server GIS chat first), runGeoExplorerGeminiTurn.ts, geoAiAgentTurn.ts; engines geoAiAgentTools.ts, geoAiCommandExecutor.ts (map commands), geoAiGisToolRunner.ts, geoAiStatsEngine.ts, geoAiLayerAttributeQuery.ts, geoAiWeatherContext.ts, geoAiBasemapQuery.ts, geoAiLiveMapContext.ts, geoAiAnalystPacks.ts, geoAiResolveMapCoords.ts, geoAiResolveTableMapLink.ts, geoAiAttachFile.ts, neighborhoodAgentPlaceIntent.ts. Providers: Gemini (geoExplorerGemini.ts, useGeminiApiKey), Claude (geoAiChatClaude.ts, useClaudeApiKey), DeepSeek (useDeepseekApiKey), Ollama (useOllamaConfig, /api/ollama/chat);

server backend/services/geoai-chat (/api/geoai-chat/*, GeoPandas/Shapely/Rasterio + optional OpenAI). Map side-effects: pin (si-geo-ai-pin), linked selection highlight (si-geo-ai-sel-*), fly-to, inspect cards.

3.14. 14. Cartography: symbology, labels, popups, legends, tables

- Symbology Studio: SiSymbologyStudioPanel, SiSymbologyClassesTable, SiPointSymbolSection, LayerProperties, LayerManager (SymbologyStyle single / unique / color / size / color_size / dot_density / threshold_markers; SymbologyClassMethod jenks / quantile / equal_interval; SymbologyColorRamp viridis / blues / greens / plasma / magma / turbo); helpers symbologyHelpers.ts, siSymbolStyleStudio.ts, siPointSymbolGallery.ts, siPolygonSymbolGallery.ts, lib/arcgisDrawingInfoMapbox.ts, arcgisPointSymbolMapbox.ts, mapboxPaintSanitize.ts.
- Labels: SiLayerLabelingPanel, lib/siLayerLabelStyle.ts.
- Popups / identify: SiLayerPopupConfigurator, LayerAttributePopupBody, FeatureIdentifyPopupCard, SiFeatureInspectPopup, MapPopup, lib/arcgisAttributeDisplay.ts.
- Legends: SiLayerLegendFloatingPanel, LayerLiveLegendPanel, LayerLiveLegendFloatingPanel, SiMapAnalysisLegendOverlay, SiCropAlertMapLegend, HydroLegendTool.
- Tables: TablePanel, GeoExplorerDynamicTable, GisSelectionAttributeTablePanel.
- Field visibility: FieldVisibilityControl.

3.15. 15. Raster georeferencing

rasterGeoreference/RasterGeoreferencePanel, RasterGeoreferenceRibbon, GeoreferenceFloatbar, useRasterGeoreferenceTool, lib/raster/siRasterTileService.ts, rasterGeorefPlacement.ts; GCP overlay si-gcp-overlay; backend /api/raster/* (georeference, COG/WMS).

3.16. 16. Hotkeys

Context	Keys	Action
Drawing	Ctrl/Cmd+Z · Ctrl/Cmd+Y / Ctrl/Cmd+Shift+Z · Esc · Backspace · Enter	undo · redo · cancel · drop vertex · commit
Selection	Shift · Ctrl · Ctrl+Shift · Esc	add · remove · from-current · close/deactivate
Measure	double-click · right-click	finish · undo vertex
MapSwipe handle	← / →	nudge split ±2 %
Go To XY	Enter	fly to coordinates
Search results	↑ ↓ Enter Esc	navigate / select / dismiss
Panels / flyouts / legends	Esc	close

4. PART III — Libraries, services, file index

4.1. Frontend libraries used by the canvas (frontend/package.json)

Library	Role
mapbox-gl ^3.18, react-map-gl ^8.1	GL canvas, globe, terrain, sources/layers/markers
maplibre-gl, maplibre-gl-layer-control	Alternate GL engine (other pages)
leaflet, react-leaflet, leaflet-draw	Legacy GIS Map (shares basemap catalog)
@turf/turf	Geometry ops, areas, buffers, centroids
proj4	CRS reprojection (rasters, WKT clip)
geotiff	GeoTIFF decode / encode
shpjs, @tmcw/togeojson, jszip, papaparse, xlsx	Vector import (SHP zip, KML/KMZ/GPX, CSV, Excel)
pmtiles, @mapbox/vector-tile, pbf	FTW Global PMTiles MVT decoding
chart.js, react-chartjs-2, chartjs-plugin-zoom	Time-series and dashboard charts
jspdf, jspdf-autotable, exceljs, react-to-print	PDF / Excel / print reports
@fortawesome/fontawesome-free	Icons
react-router-dom	Route /satellite, query params

4.2. Backend services (backend/services/*, services/*) and Node API routes

Service	Purpose	Consumer
agri-field-boundary (:8092)	Mask R-CNN / Delineate-Anything / AFD / FTW-S2 field delineation + SEN2SR	Field Boundaries, EO enrichment segmentations, Sen2SR controls
delineate-anything	YOLO field instance segmentation (FBIS)	Training & AI inference
eo-layer-enrichment	S2 zonal index enrichment CLI/API	EO Layer Enrichment
geoai-chat (:8099)	Spatial Q&A (GeoPandas, optional OpenAI)	GeoAI agents
geoai-inference (:8098)	geoai Mask R-CNN (buildings, cars, ships, solar, parking)	AI DL detection
image-classification	SLIC / Felzenszwalb + classify / accuracy	Image Classification wizard
sam-detection (:8090)	Interactive SAM	SAM Detection

Service	Purpose	Consumer
sam2-refinement (:8096)	SAM2 polygon refine	SegFormer field pipeline
segformer-detection (:8095)	SegFormer-B5 semantic segmentation	SegFormer, Training health
temporal-transformer (:8097)	Crop type on field polygons (+ Prithvi)	SegFormer temporal step
tree-detection	YOLO tree boxes	Tree Detections
services/geodash-api	Dataset upload / spatial features (GeoDash)	Not SI canvas core
Node routes	/api/terrain/esri-rgb, /api/chirps/*, /api/stac/search, /api/crop-classification/*, /api/flood-monitoring/*, /api/training/*, /api/inference/*, /api/ftw-training/*, /api/raster/*, /api/ollama/*; analysis_engine /mpc/*	Terrain, CHIRPS, STAC, Prithvi, SAR flood, training, georef, LLM proxy, MPC

External: Sentinel Hub / CDSE OGC WMS + Statistical API, Microsoft Planetary Computer STAC, Esri ArcGIS Online tiles & services, Google VT tiles, OpenStreetMap / OpenTopoMap / Carto tiles, Open-Meteo, WaPOR, CDS HiPS sky surveys.

4.3. File index (primary)

- Page & stack: `pages/satellite/SatelliteIntelligence.tsx` (MapGL children $\approx$ L24477-26028; DOM overlays $\approx$ L26030-27900), `SatelliteIntelligence.css`, `basemapCatalog.ts`, `siInstanceScope.tsx`, `useMapOverlayIsolation.ts`.
- Z-order: `lib/siMapAnalysisLayerOrder.ts`, `lib/siCustomLayerZOrder.ts`.
- WMS / AOI: `lib/sentinelHubWms{Instance,Layers,AoiClip}.ts`, `siSentinelAoiWms{Stack,FlickerFree,Imperative}.ts`, `siAoiManager.ts`, `siAoiMaskBuilder.ts`, `siAoiLayerModeClipCache.ts`, `siDrawnAoiLiveIndex.ts`.
- Terrain / globe: `lib/agroCloudMapTerrain.ts`, `siGlobeCockpit.ts`, `siGlobeSky*.ts`, `siMapboxGlobeCompat.ts`, `agroCloudMapNavigation.ts`.
- Imports: `utils/FileLoader.ts`, `lib/siCustomLayerFactory.ts`, `components/SiImportedCustomLayersOverlay.tsx`, `lib/arcgis*.ts`, `lib/raster/*`, `pages/satellite/gisDataManager/`.
- Chrome: `components/SatelliteMapAnalysisChrome.tsx`, `SatelliteContextualAnalysisDock.tsx`, `MapToolsDock.tsx`, `MapToolboxLayerList.tsx`, `MapToolboxAddGisLayerFlyout.tsx`, `SatelliteMapProcessingOptionsPortal.tsx`.
- Draw / measure / select / swipe: `SiMapDrawWidget.tsx`, `RemoteSensingDrawingToolbar.tsx`, `drawingUtils.ts`, `MeasurementPanel.tsx`, `lib/measurement/measurementEngine.ts`, `components/gisSelection/*`, `lib/gisSelection/*`, `SiMapSwipeControl.tsx`, `siMapSwipeTiles.ts`.

- Remote sensing: RemoteSensingToolboxPanel.tsx, RemoteSensingLayerSelect.tsx, lib/remoteSensing*.ts, agroCompositeIndices.ts, collectionIndexCatalog.ts, layerLiveLegendCatalog.ts, lib/remoteSensing/rsIndexGeoTiffExport.ts, lib/chirpsRainfall/*.